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7. Worked example: a complicated system input

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Example 7.1 Find the general real solution to the second order single inhomogeneous ODE

$$\ddot{x} + x = \tan t. \quad (4.153)$$

Recall from a previous problem that the companion system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{r} \quad \text{where } \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} 0 \\ \tan(t) \end{pmatrix}. \quad (4.154)$$

Use variation of parameters to find the general solution to this inhomogeneous system. Start by finding a fundamental matrix $\mathbf{X}$ for the homogeneous system

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \quad \text{where } \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (4.155)$$

The eigenvalues of $\mathbf{A}$ are $\pm i$, and the corresponding eigenvectors are $\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$. These give the normal modes

$$\begin{pmatrix} 1 \\ i \end{pmatrix} e^{it} \quad \text{and its complex conjugate.} \quad (4.156)$$

To find the real-valued normal modes, we take the real and imaginary parts of $\begin{pmatrix} 1 \\ i \end{pmatrix} e^{it}$:

$$\begin{pmatrix} 1 \\ i \end{pmatrix} e^{it} = \begin{pmatrix} 1 \\ i \end{pmatrix} (\cos t + i \sin t) = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + i \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}. \quad (4.157)$$

Therefore, a real-valued fundamental matrix is the following matrix:

$$\mathbf{X} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}. \quad (4.158)$$

Variation of parameters says that the general solution to the inhomogeneous system is $\mathbf{X} \int \mathbf{X}^{-1} \mathbf{r} dt$. Let us compute the integral:

$$\int \mathbf{X}^{-1} \mathbf{r} dt = \int \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \begin{pmatrix} 0 \\ \tan t \end{pmatrix} dt \quad (4.159)$$

$$= \int \begin{pmatrix} -\sin^2 t / \cos t \\ \sin t \end{pmatrix} dt = \begin{pmatrix} \sin t - \ln |\sec t + \tan t| \\ -\cos t \end{pmatrix} + \mathbf{c}. \quad (4.160)$$

This gives the general solution

$$\mathbf{x}(t) = \mathbf{X}\mathbf{v} = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \left[\begin{pmatrix} \sin t - \ln |\sec t + \tan t| \\ -\cos t \end{pmatrix} + \mathbf{c} \right].$$

The general solution to the original single ODE is the first component of $\mathbf{x}(t)$

$$x(t) = -\cos t (\ln |\sec t + \tan t|) + (c_1 \cos t + c_2 \sin t).$$

(Check that the second component of $\mathbf{x}(t)$ is indeed equal to $\dot{x}$).

input

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